

MTU15 effects on bids, prices and quantities critical/flat identification for ID15 and DA15

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Abstract

The two MTU15 reforms turn within-hour bidding into a strategic margin, and the dispatchable fleet exploits it. We see this directly in the bid curves: in critical hours (morning ramp and evening peak) where within-hour residual demand actually varies, ladders widen substantially — broadly under ID15 (CCGT, hydro, pump-storage) and more selectively under DA15 (hydro, CCGT). Wind, whose output spans all hours rather than concentrating in peaks, shows little critical-flat differential. The clearest quantity-side causal trace is on the day-ahead auction-cleared mix under DA15: CCGT cleared volume rises ~ 59 GWh/day net of the 2024 same-calendar placebo (joint 95% credible interval $[+43, +75]$) — the size implied by finer granularity raising CCGT's chance of being marginal across the four quarters of a peak hour. Pump-storage independently corroborates the within-day arbitrage channel: under DA15 it shifts $+237$ MWh of day-ahead generation toward critical hours, concentrating its arbitrage exactly where granularity creates value. Clearing prices move in the directions the two-market bidding contrast would predict, but their placebo-net credible intervals bracket zero; we do not claim causal price magnitudes.

1. Setup: outcomes, windows, and seasonal control

Takeaway. We measure how MTU15 reshapes three families of outcomes: per-curve bid shape; aggregate clearing price and cleared energy; and post-only within-hour dispersion. Identification compares *critical* clock-hours (those with the most within-hour load variation) to *flat* clock-hours (overnight; the within-day control) across each reform's pre/post window. Outcomes whose pre/post comparison crosses different seasons — chiefly DA15 prices and cleared quantities — are additionally seasonally adjusted.

Two reforms. The Spanish wholesale electricity market shifted from a 60-minute to a 15-minute market time unit in two steps: **ID15** (2025-03-19, intraday auctions and continuous market) and **DA15** (2025-10-01, day-ahead). After each reform there are four periods per clock-hour; before, one. The *operación reforzada* regulatory stance (REE's post-blackout reinforced operation, continuous from 2025-04-28) is a continuous confound that forces CCGT inclusion; tight pre/post windows around each reform date hold its level constant within each arm.

Variables. Each (unit u , date d , period p) sell offer is a tranche ladder $\{(p_k, q_k)\}_{k=1}^{K_{u,d,p}}$ where p_k is the bid price (EUR/MWh) and q_k the quantity (MW per OMIE *Potencia* field, `ficherosomie137.pdf` §5.1.4.2). Offers carry one of three structural types: *simple* (a bare price–quantity ladder), *minimum-income condition* (MIC), or *block* (all-or-nothing across periods); CCGT is the outlier with the bulk of its offers in MIC or block form (§4, §5(d)). Let $\text{MCP}_{d,p}$ be the OMIE clearing price *of the market in which the offer was submitted* (day-ahead offers, in the OMIE files `cab/det`; intraday-auction offers in session s , files `icab/idet`), and $h = 140$ EUR/MWh the bandwidth. Define the in-band set, MW

weights, and in-band weighted mean bid price:

$$\mathcal{B}_{u,d,p} = \{k : |p_k - \text{MCP}_{d,p}| \leq h\}, \quad w_k = \frac{q_k}{\sum_{j \in \mathcal{B}_{u,d,p}} q_j}, \quad \bar{p}_{u,d,p} = \sum_{k \in \mathcal{B}} w_k p_k.$$

Per-curve (in-band, MW-weighted; one value per (unit, date, period)):

$$\sigma_{p,u,d,p} = \sqrt{\sum_{k \in \mathcal{B}_{u,d,p}} w_k (p_k - \bar{p}_{u,d,p})^2}, \quad N_{\text{eff},u,d,p} = \left(\sum_{k \in \mathcal{B}_{u,d,p}} w_k^2 \right)^{-1}.$$

σ_p = price spread of the bid ladder within a single quarter; N_{eff} = effective tranche count of that quarter (1 = a single block, large = many equal-sized tranches).

Aggregate: $\text{MCP}_{d,p}$ is the OMIE clearing price (EUR/MWh); $Q_{d,p} = \sum_{u,k} q_{u,d,p,k}^{\text{cleared}}$ is the system DA cleared quantity per (date, period)¹.

Within-hour dispersion (post-only, used to calibrate the critical/flat partition in §2). The per-curve σ_p and N_{eff} describe the shape of a *single* quarter's ladder. Here we instead summarize how price and shape vary *across the four quarters* of a clock-hour — a direct trace of within-hour discrimination, mechanically zero under MTU60. For each (unit, date, clock-hour h) cell with all four quarters present, let $q \in \{1, 2, 3, 4\}$ index quarters, $\bar{p}_q = \sum_{k \in \mathcal{B}_q} w_k p_k$ the in-band MW-weighted mean bid price of quarter q , $\sigma_{p,q}$ and $N_{\text{eff},q}$ its per-curve metrics, $m_q = \sum_{k \in \mathcal{B}_q} q_k$ the in-band MW summed over the quarter, $\bar{m} = \frac{1}{4} \sum_q m_q$ the mean across the four quarters, and MCP_q the quarter- q clearing price:

$$D_{\text{price}} = \text{sd}_q(\bar{p}_q), \quad D_{\text{qty}} = \text{sd}_q(m_q)/\bar{m}, \quad \text{SD}_{\text{price}} = \text{sd}_q(\text{MCP}_q), \\ D_{\sigma} = \text{sd}_q(\sigma_{p,q}), \quad D_N = \text{sd}_q(N_{\text{eff},q}).$$

D_{price} , D_{qty} , SD_{price} are across-quarter shifts in bid LEVEL, bid VOLUME, and system clearing prices; D_{σ} , D_N are across-quarter shifts in ladder SPREAD and TRANCHE COUNT. All five are zero under the MTU60-equivalent strategy (identical bids in every quarter), so any positive value post-MTU15 is a direct trace of within-hour discrimination.

Windows.

Reform	Pre window	Post window	Reforzada
ID15	2024-12-19 → 2025-03-18 (3 mo)	2025-03-19 → 2025-04-27 (40 d, clean)	absent both
DA15	2025-07-01 → 2025-09-30 (3 mo)	2025-10-01 → 2025-12-31 (3 mo)	active both, constant

Method by outcome. The table below previews the design; §3 develops the formal restrictions. Bid shape uses a DiD with the critical/flat partition; clearing prices and per-tech cleared energy use a Bayesian Structural Time Series counterfactual (Brodersen et al. 2015, applied to electricity reform by Märkle-Huß et al. 2017). The BSTS pre-window for DA15 starts after the post-blackout reforzada onset (2025-04-28), holding the regulatory regime constant; ID15's post-window is pre-blackout. Both reforms have a placebo on the same calendar window of 2024 to falsify residual cross-season trend.

¹The OMIE assigned-power field is per-period instantaneous power (MW), invariant to period length, so one 15-min quarter and one 60-min hour at the same MW are directly comparable as rates. Aggregating MW across the four quarters of a clock-hour gives the stock of energy delivered in that clock-hour, $\text{MWh}_h = \sum_{q \in h} \text{MW}_q \Delta t_q$. Aggregate-quantity coefficients below are reported as MWh per clock-hour (numerically equal to the per-period mean MW because the integration is over 1 h, but the unit is energy, not power).

Outcome	Method	Granularity	Pre / Post window	Identifies
σ_p , N_{eff} (bid shape)	DiD crit–flat	per-curve	3 mo / 3 mo	critical–flat differential
DA price, cleared energy per tech	BSTS	daily	156 d / 40 d (reforzada-constant)	DA15
IDA price, cleared energy per tech	BSTS	daily	278 d / 40 d (post-IDA-reform)	ID15
Placebo 2024 (each reform)	BSTS	daily	same windows shifted to 2024	cross-season artefact

Seasonal control. Bid-shape outcomes need none: σ_p and N_{eff} are MCP-centered and MW-weighted on the in-band ladder, so summer→winter price-level seasonality does not transmit; the flat-hour control absorbs common-to-all-hours bid-shape seasonality and the placebo (§5(b)) checks for pre-existing trends in the critical–flat differential. For *level outcomes* (prices, cleared energy), seasonality is controlled by the BSTS covariates (wind, solar, gas) plus a weekly seasonal cycle and the local-linear trend (§3.B), and residual cross-season trend is gauged by the 2024 same-calendar placebo. For the supporting pump-storage DiD (§4/5(c)) we use a same-calendar-month restriction (pre Oct–Dec 2024, post Oct–Dec 2025).

2. The critical/flat partition: where granular pricing has economic content

Takeaway. Within-hour discrimination has economic content only where within-hour residual demand actually varies. Critical hours (morning ramp + evening peak) are where MTU15 can bite; flat hours (overnight) are the within-day control. Post-MTU15 within-hour dispersion is much larger in critical than flat hours on every bid-level and bid-shape outcome we measure — the partition correctly identifies where the reform lives.

Intuition. Granular pricing only has economic content where within-hour residual demand² actually varies; if a clock-hour is internally flat, the four quarters look the same and there is nothing to discriminate on. We measure within-hour variation by $\sigma_{\text{within}}(h, d) = \text{sd}_q[\text{load} - \text{wind} - \text{solar}]_{q,h,d}$ on 15-min residual-demand data³ and partition clock-hours by median σ_{within} :

- **Critical** $\mathcal{C} = \{5, \dots, 8, 16, \dots, 22\}$: morning ramp + evening peak; $\sigma_{\text{within}} \in [522, 1303]$ MW.
- **Flat** $\mathcal{F} = \{1, 2, 3\}$: overnight; $\sigma_{\text{within}} \in [336, 357]$ MW.
- **Midday** $\{11, \dots, 14\}$ (solar-marginal, $\sigma_{\text{within}} \in [468, 502]$ MW): excluded from main specs; kept for falsification (§5(a)).

Formal calibration of the partition (post-only). Let $D_{d,h}$ stand for any one of the five within-hour dispersion outcomes (D_{price} , D_{qty} , SD_{price} , D_{σ} , D_N) on date d , clock-hour h . Under MTU60 the within-hour quarter dimension does not exist, so $D_{d,h}$ is mechanically zero pre-reform; we use the post-MTU15 sample to *describe* where within-hour dispersion lives across hour-classes. A date-FE

²Spanish-market convention calls this the *hueco térmico* (thermal gap) — the load that thermal and other dispatchable generation must cover after subtracting wind and solar.

³ENTSO-E Transparency Platform: A65 (actual total load) and the wind/solar component of A75 (actual generation per type, PSR codes B16, B18, B19); window 2023-01-01 to 2026-05-14.

regression is a convenient way to summarise the critical–flat differential:

$$D_{d,h} = \delta_d + \theta \mathbf{1}\{h \in \mathcal{C}\} + \varepsilon,$$

fit per outcome. θ is the post-MTU15 critical–flat dispersion gap. This is a descriptive contrast, not a causal estimate: operational ramp constraints in critical hours can mechanically generate within-hour SD, and the partition calibration cannot separate operational from strategic dispersion.

Outcome	Subsample	ID15	DA15	crit/flat ratio (IDA, DA)
D_{price} (EUR/MWh)	per unit (pooled)	0.39*** (0.05)	1.35*** (0.11)	4.1×, 7.4×
D_{qty} (CV)	per unit (pooled)	0.024*** (0.003)	0.033*** (0.002)	4.3×, 4.9×
SD_{price} (EUR/MWh)	system	4.55*** (0.40)	3.97*** (0.37)	2.8×, 2.6×
D_{σ} (EUR/MWh)	CCGT	0.42*** (0.07)	1.95*** (0.15)	3.2×, 40.7×
	Hydro	0.34*** (0.04)	0.33*** (0.05)	4.0×, 2.2×
	Hydro pump	0.077*** (0.02)	−0.002 (0.01) ns	64×, 1.0×
	Wind	−0.032 (0.05) ns	0.10*** (0.01)	0.9×, 2.0×

SE clustered by date; *** $p < 0.01$, ns = not significant. Pooled per-unit and system rows show the partition has economic content on both bid-level (D_{price} , D_{qty} , SD_{price}) and bid-shape (D_{σ}). The per-tech breakdown of D_{σ} shows where within-hour discrimination is concentrated: CCGT exhibits by far the largest critical–flat differential under DA15, as one would expect from the marginal scarcity-tier technology; hydro is also significant; pump-storage shows none under DA15, because its strategic margin is day-ahead quantity allocation rather than within-hour ladder shaping — the same logic that drives the pump-storage finding in §4. Wind is small under DA15 and indistinguishable from zero under ID15, matching the fact that ~84% of its in-band curves are a single block.

3. Identification: DiD for bid shape, BSTS for prices and quantities

Takeaway. Bid shape is identified by a critical/flat DiD: the (critical – flat) shift across each cutover is the ATT under parallel trends. Clearing prices and per-tech cleared energy are identified by a Bayesian Structural Time Series counterfactual on a daily panel: conditional on the wind and solar renewable-supply covariates, the post-cutover series departs from the model’s pre-trained extrapolation by the treatment effect; a 2024 same-calendar placebo gauges residual cross-season trend.

u unit; d date; p period; h clock-hour; s IDA session; T_r reform date. Treatment status $D_{d,h} \equiv \mathbf{1}\{d \geq T_r\} \cdot \mathbf{1}\{h \in \mathcal{C}\}$; sample restricted to $h \in \mathcal{C} \cup \mathcal{F}$ for the DiD.

A. Per-curve DiD for bid shape ($Y \in \{\sigma_p, N_{\text{eff}}\}$, run per-tech):

$$Y_{u,d,p[s]} = \alpha_u + \beta \mathbf{1}\{d \geq T_r\} + \xi \mathbf{1}\{h \in \mathcal{C}\} + \theta D_{d,h} + \varepsilon.$$

Under the parallel-trends assumption $E[Y(0) \mid \mathcal{C}, d \geq T_r] - E[Y(0) \mid \mathcal{C}, d < T_r] = E[Y(0) \mid \mathcal{F}, d \geq T_r] - E[Y(0) \mid \mathcal{F}, d < T_r]$, the DiD coefficient θ identifies the ATT on the critical-hour bid shape. Flat hours act as a low-exposure control (mechanically reachable post-cutover but with negligible within-hour residual demand variance, §2), giving the standard 2×2 DiD without staggered-treatment negative weights (de Chaisemartin & D’Haultfoeulle, 2026, ch. 3). SEs are clustered by date.

B. Bayesian Structural Time Series for prices and per-tech cleared energy. Following Brodersen et al. (2015) as applied to the EPEX 15-min reform by Märkle-Huß et al. (2017), the daily response y_t (DA or IDA clearing price; per-tech daily cleared GWh) decomposes as

$$y_t = \mu_t + \gamma_t + X_t' \beta + \varepsilon_t,$$

where μ_t is a local linear trend with stochastic slope (default priors), γ_t is a 7-period weekly seasonal cycle, $X_t = (\text{wind GWh}, \text{solar GWh}, \text{gas EUR})$ are exogenous renewable-supply and fuel-cost covariates, and β is a spike-and-slab regression coefficient. Märkle-Huß et al. used only wind and solar for EPEX; we add gas because Spain has meaningfully different gas exposure than Germany did in their window. Demand is excluded because it is itself partly an outcome and would induce simultaneity. Pre-cutover data train the model; the counterfactual at $t \geq T_r$ is $\hat{y}_t(0) \equiv E[y_t(0) \mid X_t, t < T_r]$, and the treatment effect at t is $y_t - \hat{y}_t(0)$. The post-window average effect and its 95% credible interval come directly from the Bayesian posterior.

Window choice follows the principle of holding the relevant regulatory regime constant across pre and post. For DA15 the maximum reformed-constant pre-window is 2025-04-28 $\rightarrow$ 2025-09-30 (156 days, starting at reformed onset). For ID15 the maximum regime-constant pre is 2024-06-14 $\rightarrow$ 2025-03-18 (278 days, post-IDA-reform 3-session architecture, includes the December 2024 ISP15 introduction whose effect on the daily-mean price is small and is absorbed by the regression coefficients on wind, solar and gas). Märkle-Huß et al.’s shorter 80-day pre is reported as a robustness check (§5(c)); it gives larger magnitudes because the local-linear trend cannot extrapolate cross-season. Two placebos — one on the same calendar 2024 for each reform — falsify residual cross-season trend; placebo-net effects are the cleanest identification of the treatment.

The identifying restriction: conditional on X_t , the unobserved $y_t(0)$ for $t \geq T_r$ equals the pre-trained model’s extrapolation. This rules out any unobserved level shift coincident with the cutover that is not predicted by X_t or by the local-trend extrapolation, and the placebo gauges whether the placebo-window extrapolation is itself biased by seasonal trend.

4. Results: bid-shape widening, day-ahead CCGT auction-cleared scale-up, and directional price movements

Tech selection. Spec A is run on the four technologies that bid into the in-band region with non-trivial frequency: CCGT (the focal scarcity-tier technology), hydro and hydro pump (cleanest from an omitted-variable-bias standpoint — their marginal-cost stacks are insensitive to the gas-renewable mix), and wind (the dominant sell-side price-setter, included for consistency with Brown & Reguant 2026). Solar PV is excluded because it bids effectively zero in-band scarcity-tier capacity; nuclear is excluded because its dispatch is predominantly bilateral.

Per-tech caveats. Wind within-hour σ_p mixes sub-hourly forecast revisions with strategic ladder shaping, so any positive wind DiD reads as a price-taking baseline (see table notes). The offer-type composition concern (Figure 1, §5(d)) is CCGT-specific: hydro, hydro pump and wind are $\sim 100\%$ simple offers in both arms, so their Spec A DiDs below are simple-only by construction.

Outcome (DiD θ)	Subsample	ID15	DA15
σ_p (EUR/MWh)	CCGT	2.45*** (0.58)	1.30* (0.75)
	Hydro	1.17*** (0.44)	1.69*** (0.29)
	Hydro pump	0.25 (0.28)	-0.63* (0.36)
	Wind [†]	0.32*** (0.10)	0.17*** (0.04)
N_{eff}	CCGT	0.79*** (0.30)	0.44*** (0.06)
	Hydro	0.21*** (0.06)	-0.04 (0.04)
	Hydro pump	0.18** (0.08)	-0.05 (0.03)
	Wind [†]	0.017** (0.007)	0.009*** (0.003)

Notes: SEs in parentheses, clustered by date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Per-tech subsample regressions, unit FE absorbed via within-unit demeaning. [†]Wind DiDs read as a price-taking baseline: 84% of in-band wind sell curves are a single block, with critical and flat scarcity slopes indistinguishable; magnitudes are 5–10 $\times$ smaller than CCGT. *DA15 CCGT σ_p by offer type* (Figure 1): simple-only +3.53*** ($t = 3.60$), MIC -0.90 (ns), block +0.29 (ns); N_{eff} widens broadly across offer types.

BSTS results for prices and per-tech auction-cleared energy.

Outcome	Tech	ID15		DA15	
		Effect	Placebo	Effect	Placebo
Clearing price (EUR/MWh)	—	-38.34***	-18.08*	+2.97	-8.54
Cleared (GWh/day)	CCGT	-8.08***	-9.18***	+40.31***	-18.69***
	Hydro	-1.47***	+5.94***	+9.18***	+10.83***
	Wind	+5.34***	+7.99***	+13.16***	+2.15
	Solar	+6.70***	+2.55***	-3.37	+1.38
	Nuclear	+11.51***	+19.06***	-18.21**	-34.77***

Bayesian posterior point estimates; tail-probability stars * < 0.10 , ** < 0.05 , *** < 0.01 . Pre-windows respect the relevant regulatory regime constancy: ID15 pre 2024-06-14 $\rightarrow$ 2025-03-18 (278 days, post-IDA-reform 3-session regime), post 2025-03-19 $\rightarrow$ 2025-04-27 (40 days, pre-blackout); DA15 pre 2025-04-28 $\rightarrow$ 2025-09-30 (156 days, post-reforzada-onset), post 2025-10-01 $\rightarrow$ 2025-11-09 (40 days). Each *Placebo* column uses the same calendar window of 2024 (no reform; pure seasonal trend benchmark). The placebo-net contrast (real - placebo) is the identified treatment effect; we compute its joint 95% credible interval treating the two BSTS posteriors as independent (different years and model fits) and propagating their posterior variances. Only one BSTS finding survives at joint 95%: **DA15 CCGT cleared scale-up**, +59 GWh/day net (joint CI [+43, +75]). The other placebo-net contrasts — prices (IDA, DA), hydro, solar, wind, nuclear cleared — have joint CIs that bracket zero. The window-length sensitivity in §5(c) reports the Märkle-Huß 80-day-pre version of the same BSTS for comparison.

What survives. (i) **Bid-shape widening in critical hours** (Figure 1). The granularity effect is robust across both reforms. ID15 widens broadly (CCGT, hydro, and pump-storage all show positive σ_p DiDs; cross-border control attenuates by only 8%, §5(d)). DA15 is more selective: hydro σ_p and CCGT N_{eff} (broad across offer types) have clean placebos. The pooled DA15 CCGT σ_p is diluted by an offer-type composition shift; the cleaner simple-offer-only estimate is +3.53*** (see §5(d), table notes, and right panel of Figure 1). Wind DiDs are small and read as a price-taking baseline.

Per-tech σ_p DiD: bid-shape widens in critical hours across techs and reforms

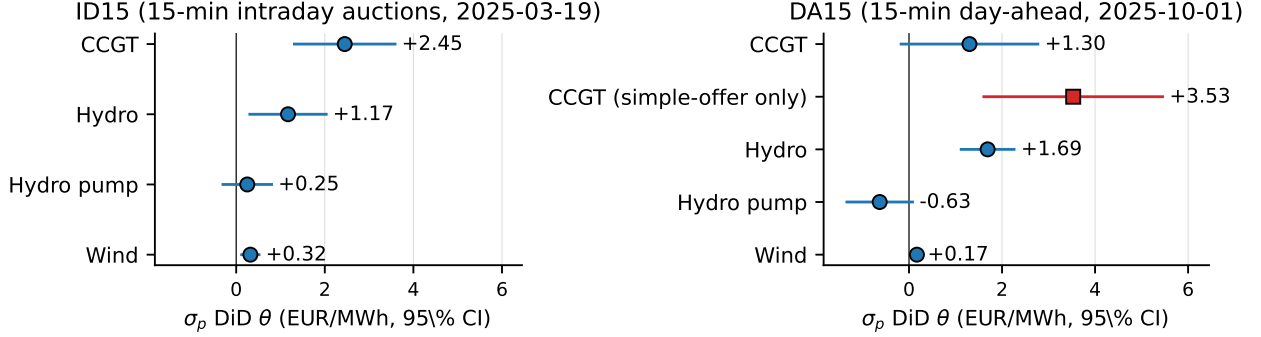


Figure 1: Per-technology σ_p DiD with 95% confidence intervals. Bid-shape widening in critical hours is positive and significant across most technology–reform combinations. *Right panel*: the DA15 CCGT pooled estimate (+1.30) is diluted by an offer-type composition shift (§5(d)); the cleaner simple-offer-only estimate is +3.53 ($t = 3.60$).

(ii) **Day-ahead CCGT auction-cleared volume scales up under DA15.** Placebo-net CCGT cleared energy rises +59 GWh/day (joint 95% credible interval [+43, +75], propagating real and placebo posterior variance) — the same channel as the bid-shape widening in (i): MTU15-DA lets CCGT discriminate finely within the hour, raising its expected acceptance probability across the four quarters of a peak hour. Hydro, solar and nuclear are essentially null once the 2024 same-calendar placebo is netted off (joint CIs bracket zero). Wind cleared also rises (+11 GWh/day net, joint CI [−1, +23]); the effect is borderline and wind shows little critical-flat differential in §2 (its output spans all hours rather than concentrating in peaks), so the granularity channel does not naturally explain it — the +11 reads as residual cross-year dispatch variation the placebo does not absorb.

(iii) **Pump-storage within-day arbitrage.** Critical-clock-hour DA generation rises +237 MWh under same-calendar ($t = 5.0$; baseline cross-season +634 is inflated by summer/winter demand-mix). Under same-calendar, generation scales up in both hour-classes but disproportionately in critical: flat 117 → 259 MWh, critical 645 → 1018 MWh; the pumping pattern stays concentrated in flat hours pre and post. The technology with the largest within-hour arbitrage value scaled up its critical-hour energy once it could bid quarter by quarter.

(iv) **Prices move directionally but joint credible intervals bracket zero.** Placebo-net point estimates are −20 EUR/MWh for IDA after ID15 (joint 95% CI [−50, +9]) and +12 EUR/MWh for DA after DA15 (joint CI [−15, +38]). The signs track the two-market bidding contrast (§6): CCGT bids at marginal cost (~EUR 82–96/MWh) in IDA with ~0% withheld, so finer granularity lets cheaper marginal suppliers undercut in some within-hour quarters and drags the IDA average down; CCGT’s two-tier scarcity-aware DA strategy means the auction-cleared mix shifts toward CCGT under DA15, raising the day-ahead level at the margin. The joint credible intervals (propagating the placebo’s own posterior variance) bracket zero, however, so these price magnitudes do not stand as causal estimates — the BSTS posterior for each side is individually significant, but the placebo-netted contrast is not.

5. Robustness: surviving findings hold; price effects do not

Takeaway. Pump-storage and bid-shape findings survive the standard threats; the BSTS DA15 CCGT cleared scale-up survives at joint 95%; price effects do not.

(a) Hour-class definition does not affect our results. Three checks. *Midday-vs-flat falsification:* σ_p DiD is $\sim 65\%$ of the critical-vs-flat magnitude — dose-response in within-hour variance, not a placebo failure. *Weekday-only:* at the baseline. *Extended flat set* (add hour 0, then hour 23): pooled σ_p survives every extension.

(b) ID15 results are window-invariant; DA15 CCGT σ_p is the one window-sensitive finding. *Extended ID15 pre window* (back to 2024-06-14): σ_p attenuates but the sign and significance survive. *Pre-only placebo* (fake T_r at the midpoint of each pre window): ID15 is clean, while the DA15 σ_p placebo flags a pre-existing critical-flat trend — so the DA15 bid-shape claim rests on hydro σ_p and CCGT N_{eff} , where the placebo is clean.

(c) Seasonal control: BSTS placebo for level outcomes, same-calendar DiD for pump-storage; window-length sensitivity. The BSTS placebo (§4 table) handles cross-season trend for prices and per-tech daily cleared energy: the placebo coefficient is the seasonal component and the placebo-net effect is the treatment. Pre-windows are chosen to hold the relevant regulatory regime constant (DA15: 156 d post-reforzada-onset; ID15: 278 d post-IDA-reform). Märkle-Huß et al.’s shorter 80-day pre gives substantially larger raw magnitudes (e.g. ID15 IDA price -62 vs -38 long pre; DA15 nuclear -67 vs -18) but rests on a winter-only or summer-only pre, so the local-linear trend extrapolates a cross-season counterfactual blindly; we report the 80-day result as a window-length robustness check, the long pre as the headline. We also add gas as a covariate beyond the Märkle-Huß wind+solar set because Spain has meaningfully different gas exposure than Germany. The pump-storage finding rests on a separate DiD with same-calendar restriction (pre = Oct–Dec 2024, post = Oct–Dec 2025), retaining $+237^{***}$ MWh ($t = 5.0$) under same-calendar with weather controls, corroborated by a Fourier-SA cross-check ($+350$, $t = 7.4$).

(d) None of the four candidate confound channels overturns the surviving findings. *Cross-border flows for ID15:* the same ID15 cutover moved pan-European IDA SIDC and XBID to 15 min, so this is a sibling treatment, not omitted-variable bias on $Y(0)$. The cross-border critical-flat DiD⁴ is $+8965$ MWh ($t = +5.12$) for ID15 vs $+793$ ns for DA15; adding cross-border as a per-curve control attenuates ID15 σ_p by only 8% — bid-granularity dominates. *Reforzada* is absent from ID15 by construction and continuous over the DA15 cutover; its empirical signature is the same-cal DA15 CCGT cleared-energy decrease ($-1,363$ MWh), and the CCGT scarcity-tier split confirms it (withholding share $+3.7\text{pp}^{***}$, competing-tier $N_{\text{eff}} +0.18^{***}$ — reforzada parks capacity in the scarcity tier, MTU15 makes the competing-tier ladder more granular). *CCGT offer-type composition* (§4, Figure 1) understates the pooled σ_p DiD; simple-offer-only $+3.53^{***}$ is the cleaner estimate, N_{eff} widens broadly. *Per-CCGT-firm split:* the DA15 GN headline $+35.0$ is a flat-baseline collapse artefact (mean $40.7 \rightarrow 7.6$), not a within-hour MTU15 effect.

⁴ESIOS indicators 535 (FR imp), 536 (FR exp), 539 (PT imp), 540 (PT exp); net = (FR_imp + PT_imp) – (FR_exp + PT_exp).

6. Two market-structure facts that frame the bidding evidence

The two-tier CCGT bid curve and the day-ahead/intraday contrast. Every Big-4 CCGT fleet runs a two-tier day-ahead offer: a low *competing* tier near marginal cost ($\sim$ EUR 100–110/MWh) and a high *scarcity* tier above the $\sim$ EUR 200 clearing-price ceiling — shares above the ceiling: Naturgy 75%, Endesa 22%, Iberdrola 22%, EDP-HC 19%. Withholding is firm-wide, not a Naturgy story. The same fleets bid a single competitive tier at EUR 82–96/MWh with $\sim$ 0% withheld in the intraday auction, after Fase I has cleared. A unit’s marginal cost does not change between two auctions hours apart. The parked day-ahead capacity is recalled in Fase I and paid pay-as-bid through a *separate* restriction offer (procedure PO 3.2, BOE-A-2025-5342; settlement PO 14.4 ap. 20.1, BOE-A-2025-13076); the day-ahead two-tier curve is the geometry of the Fase I-recall game, not cost-reflective bidding. MTU15-DA grants CCGT a within-hour degree of freedom over its competing tier. Through this lens, the +59 GWh/day CCGT auction-cleared scale-up in §4(ii) is the competing tier expanding into the four quarters of each peak hour.

Market concentration migrated from wholesale to ancillary services. Wholesale-market HHI hovers around 1 100 over 2016–2024 (below the moderate-concentration threshold of 1 500) and falls mid-day when solar peaks (solar is less concentrated than the Big-3). Day-ahead restrictions upward HHI above 4 000; deviations upward above 6 000; tertiary upward around 5 000. Median of 4 firms with accepted upward restrictions bids. As renewables eroded wholesale-market power, technical and locational requirements concentrated the ancillary markets that absorb their variability.

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